

AI-Enhanced Decision-Making in Childhood Obesity Interventions

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Abstract. Childhood obesity represents a global public health challenge. This study employs artificial intelligence (AI) to enhance decision-making in pediatric interventions against obesity. A predictive model was developed using data from children participating in an obesity intervention program at the Pediatric Service of the Elx-Crevillent Health Department to identify which type of patient achieves a successful outcome after a standard obesity intervention. The results demonstrate the potential of AI to improve personalized healthcare and optimize resources in combating childhood obesity, specifically in defining which patients may benefit from standard procedures and which patients may require additional resources to achieve improvements. This pilot research contributes to a deeper understanding of the factors related to childhood obesity and lays the groundwork for future AI-driven innovations in pediatric health.

Keywords: Childhood Obesity · Machine Learning · Decision Making.

1 Introduction

Childhood obesity represents one of the most pressing challenges for public health today, with alarming rates continuing to rise worldwide. This health issue has reached worrying proportions in Spain [1], as in many other countries, with recent studies indicating a significant prevalence among children of various ages [2]. This condition entails immediate health risks for children and projects a considerable economic and healthcare burden for the future [3].

Currently, various studies are addressing the prediction of childhood obesity using innovative approaches and longitudinal data. Gupta et al. [4] propose a method based on the analysis of electronic health records (EHR) using recurrent neural networks (RNN) with long short-term memory (LSTM) cells, allowing for early detection of children at risk of obesity. Gou et al. [5] evaluate the accuracy of existing prediction models, emphasizing the importance of lifestyle factors in childhood obesity. Butler et al. [6] focus on developing a model to predict obesity

in New Zealand children. Fu et al. [7] employ machine learning techniques to identify risk factors in premature newborns. However, unlike these studies focusing on predicting childhood obesity based on medical history, our approach aims to predict the success of obesity interventions in pediatric patients.

This study addresses various aspects of childhood obesity management to enhance the effectiveness of interventions and decision-making in the pediatric field [8]. To achieve this, a predictive model develops to anticipate success in standard childhood obesity intervention programs by identifying the factors influencing treatment outcomes. Additionally, an examination was conducted on the factors associated with childhood obesity and their impact on intervention effectiveness. This approach aims to contribute to more precise and personalized care for children affected by obesity by identifying which patient profile may succeed with standard treatment and which profile may require a more personalized and rigorous approach to achieve satisfactory outcomes. A comprehensive evaluation of diverse artificial intelligence techniques, with a predominant focus on machine learning algorithms, was undertaken to determine their efficacy in improving decision-making in pediatric care. Lastly, an investigation is carried out into how consensus models can integrate predictions generated by multiple machine learning algorithms, aiming to provide more reliable estimates of treatment outcomes, thereby contributing to more precise and personalized healthcare for children affected by obesity.

The article follows a structured format: Section 1, the introduction provides the study's context and establishes its objectives. Section 2 focuses on data acquisition, detailing the data preparation and study design techniques. Section 3 describes the methodology, including the classification method employed and the use of the Polisabio Advanced Classifier Optimizer. The study findings are presented in Section 4, where the performance of the classification model is evaluated, and consensus models are discussed. Finally, in Section 5, the conclusions are summarized, highlighting the implications of the results and the potential of artificial intelligence in childhood obesity treatment.

2 Data Processing

This article details the tasks carried out in this preparatory action, as well as the design of the retrospective observational and multicenter study that was conducted. The study population consisted of 96 children aged between 5 and 17 years who participated in an obesity intervention program at the Pediatric Service of the Elx-Crevillent Health Department.

Hospital professionals created a data collection book, combining a literature review and retrospective study to develop predictive models in childhood obesity. Strict ethical measures were applied, preserving patient confidentiality, obtaining informed consent, and following established ethical guidelines.

In Figure 1, the 44 variables included in the study are grouped according to their typology. Additionally, the data type of each variable is detailed. In order to simplify data management, categorical variables belonging to sociodemographic

variables were encoded as numeric through a data mapping process [9]. However, the nature of these data was considered when creating the models explained in Section 3. It's noteworthy that the dataset is complete, without any missing data.

All patients receive a standard treatment for combating obesity. This treatment, offered by the Health Department, comprises improvements in diet, increased hours of exercise outside of school, and better rest time. This treatment is not specific to individual patients, as all patients receive the same information and guidelines to follow.

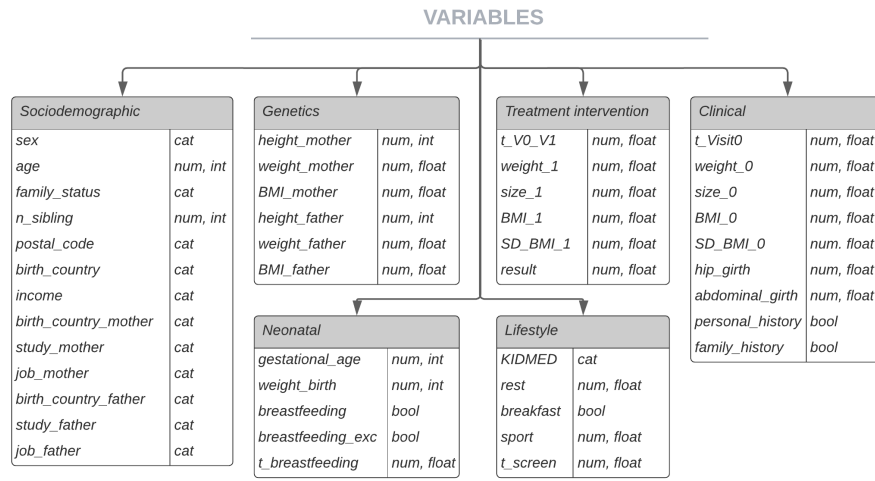


Fig. 1. Groups of variables defined in the project.

3 Methodology

This section introduces a classification model to forecast treatment success in standard childhood obesity intervention programs. Using a uniform treatment approach, the model seeks to determine whether a patient achieves a positive outcome.

Patient classification variables are defined during the initial patient visit and encompass sociodemographic, genetic, neonatal, lifestyle, and initial clinical data. Leveraging these variables, patients are categorized into two groups: those with success in standard treatment outcomes and those without success.

The model is trained through historical data analysis, learning from previous outcomes. Once the training process is completed, the model can make predictions about the outcomes of new patients, whose data are stored in a designated Excel file for prediction. This anticipatory capability facilitates informed clinical

decision-making, as it can identify which type of patient requires more specific or differentiated treatments to achieve favorable results.

This approach anticipates new patients' treatment outcomes and provides opportunities for healthcare personalization and optimization. By basing predictions on predefined variables, the model contributes to more precise and efficient healthcare delivery aligned with each patient's specific characteristics.

3.1 Classificatory Variable

The variable used to predict the treatment success in new patients is based on the treatment outcome. Given the continuous growth of children, Body Mass Index (BMI) is not a directly applicable indicator. Thus, sex-differentiated tables considering age are employed to evaluate BMI in this demographic group. Specifically, the BMI (BMI-SDS) standard deviation is utilized. This metric quantifies the disparity between the observed BMI of the patient and the expected BMI for someone of the same age and sex [10]. Calculating this discrepancy requires information on sex, age, weight, and height. Utilizing resources such as the Spanish Society of Pediatric Gastroenterology, Hepatology, and Nutrition website ensures that comparisons are made regarding specific norms for the pediatric population in Spain based on WHO guidelines from 2006/2007 [11].

Our study measured BMI-SDS at the initial visit (BMI-SDS_0) and at the visit following treatment (BMI-SDS_1). Treatment outcome is the difference between BMI-SDS at the visit after starting treatment and BMI-SDS at the initial visit (Equation 1). A negative result indicates an improvement in BMI standard deviation.

$$Result = (BMI - SDS_1) - (BMI - SDS_0) \quad (1)$$

Following the analysis conducted in the study [12] on the effects of reducing the BMI-SDS on cardiovascular risk factors, it was decided to establish a threshold of -0.15 in the outcome variable to define treatment success. This threshold was adjusted explicitly for the 6-month treatment period based on the findings provided by the aforementioned research. All results below this value indicate a more significant decrease in the BMI-SDS, representing treatment success, while values above mean treatment failure.

The variables that most significantly influence the outcome variable are: BMI-SDS_0 (Standard Deviation of the Body Mass Index), t_V0_V1 (time between doctor appointments) and BMI_mother (Body Mass Index of the patient's mother).

3.2 Polisabio Advanced Classifier Optimizer

The applied classification model is distinguished by its comprehensiveness and is highlighted as a robust tool, even within the context of a relatively small dataset comprising only 96 observations. Despite its limited size, this model has been strategically enhanced to improve its performance and enable the machine

learning algorithms employed to yield highly accurate results. Figure 2 presents a flowchart depicting a summary of the classification method process.

The main distinguishing feature in data treatment is the contribution of the “Polisabio Advanced Classifier Optimizer” classification model, which addresses the challenge of working with small datasets by applying data oversampling and balancing techniques, using Python as a programming language due to its great popularity and efficiency for data science [13]. In this section, the model and the proposed characteristics are described.

The model requires two input datasets: historical data and prediction data. In this study, historical data refers to the database of pediatric obesity treatment patients. For this study, only the variables obtained before starting treatment (initial variables) and the treatment outcome (Section before) are necessary. In the prediction data, we have the initial variables of a new set of patients for whom the treatment outcome is unknown. This model aims to predict the standard treatment outcome for new patients.

Firstly, it must be checked that the columns of both data input sets are the same because the prediction data set must have the same variables as the data used to train the model, historical data. If this is not the case, the model would return an error, and the input would have to be corrected.

Based on the continuous variable “Result” and the defined threshold for classifying patients, we derive the variable “Outcome,” which is the target variable for prediction. In this variable, patients is assigned a value of 0 (True) if they achieve success and a value of 1 (False) if they do not. So, if a patient is labeled as True, then it means that the regular treatment is working, otherwise, if labeled as False, then the patient should follow other type of treatment.

The next step is to separate the target variable “y” (Outcome) from the features “X,” and the historical data set is divided into a training set (X_train, y_train) and a test set (X_test, y_test). One-Hot encoding is used for binary encoding of categorical variables, making it easier to include them in subsequent Machine Learning models [14].

If there is data class imbalance in the target classes, stratified sampling of X_train is used to ensure that all classes of the target variable y_train are equally represented [15].

Additionally, two oversampling strategies, including RandomOverSampler [16] and SMOTE [17], are employed to generate duplicate and synthetic data, respectively, aiming to address the limitation in the database size and thus generate more individuals.

The model uses the following predictive models: LogReg [18], decision trees [19], SVM [20], Naive Bayes [21], Random Forest [22], Ada Boost [23], and XG-Boost [24]. The best samplers and oversampling ratios are selected from these models, generating y_pool, representing the best model found, in this case, it was XGB using SMOTE as a sampler and using oversampling with ratio = 3.

Algorithm 1 outlines the main components of the fine tuning of the machine learning (ML) models, which uses different criteria to refine the model and obtain the best predictions. The procedure receives two inputs, the base

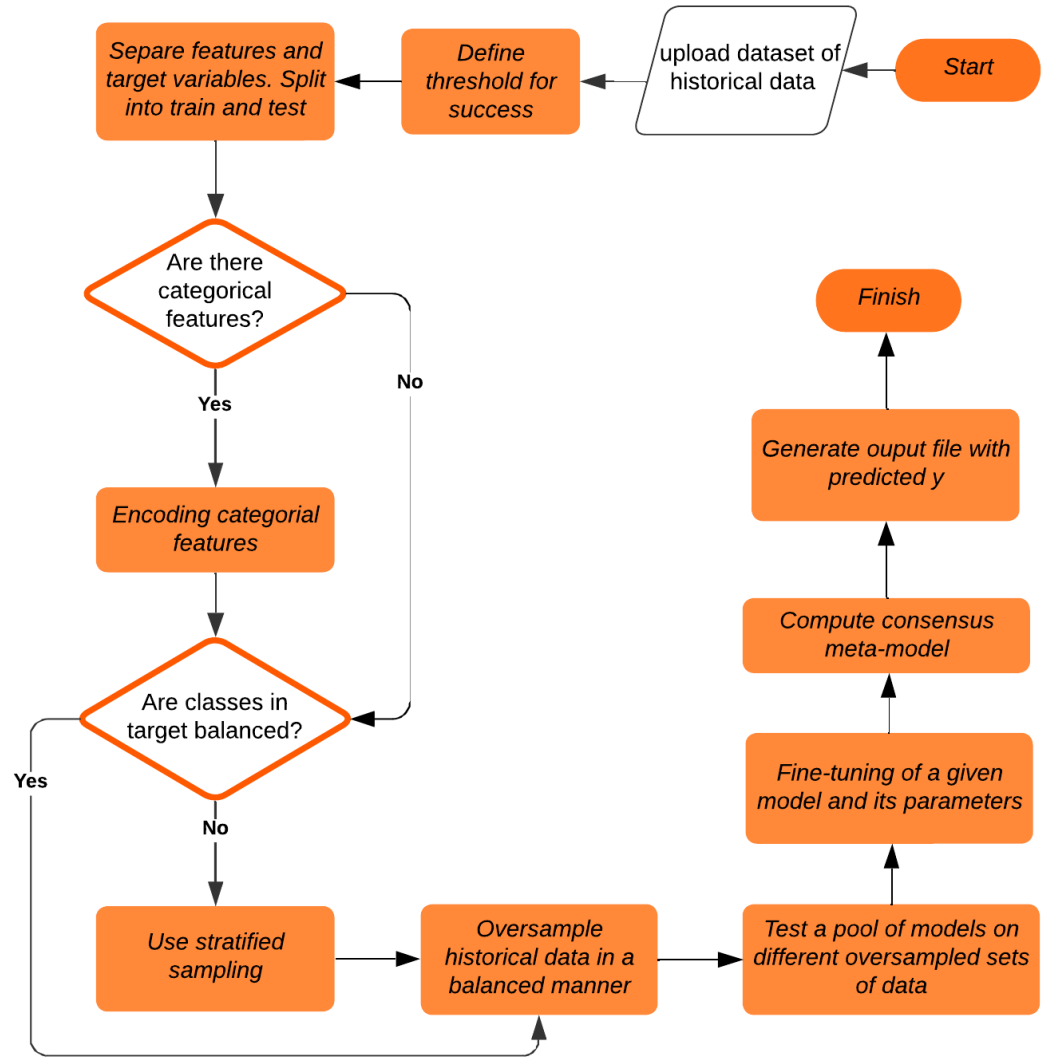


Fig. 2. Flowchart of Polisabio Advanced Classifier Optimizer.

Algorithm 1 Fine Tune ML Model**Require:** A base model b_model , a parameter grid p_grid **Ensure:** Best test score, best predictions

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    ▷ Find the best model according to the best train score
1:  $train\_s\_t, test\_s\_t, test\_pred\_t \leftarrow \text{bestTrainScore}(b\_model, p\_grid)$ 
2:  $best\_train\_s, best\_test\_s, best\_test\_pred \leftarrow train\_s\_t, test\_s\_t, test\_pred\_t$ 
    ▷ Find the best model according to the best validation score
3:  $train\_s\_v, test\_s\_v, test\_pred\_v \leftarrow \text{bestValScore}(b\_model, p\_grid)$ 
4: if  $(test\_s\_v > best\_test\_s) \wedge (train\_s\_v \geq test\_s\_v)$  then
5:    $best\_train\_s, best\_test\_s, best\_test\_pred \leftarrow train\_s\_v, test\_s\_v, test\_pred\_v$ 
6: end if
    ▷ Find the best model according to the best test score
7:  $train\_s\_e, test\_s\_e, test\_pred\_e \leftarrow \text{bestTestScore}(b\_model, p\_grid)$ 
8: if  $(test\_s\_e > best\_test\_s) \wedge (train\_s\_e \geq test\_s\_e)$  then
9:    $best\_train\_s, best\_test\_s, best\_test\_pred \leftarrow train\_s\_e, test\_s\_e, test\_pred\_e$ 
10: end if
11: return  $best\_test\_s, best\_test\_pred$ 

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ML model (b_model), and a selection of parameters to test (p_grid). First, using standard cross-validation techniques, the `bestTrainScore` procedure computes the model with the best training score. The procedure returns three values, the model training accuracy score ($train_s_t$), the test accuracy score ($test_s_t$), and the predictions from the test set ($test_pred_t$). These values are assigned to be the best training accuracy score ($best_train_s$), best test accuracy score ($best_test_s$), and best predictions ($best_test_pred$) found so far. Next, the `bestValScore` procedure computes the model with the best validation score. The procedure splits the training set into two sets, reduced training and validation sets, and employs cross-validation techniques with the validation set to fine tune the model. The procedure returns three values, the model training accuracy score ($train_s_v$), the test accuracy score ($test_s_v$), and the predictions from the test set ($test_pred_v$). Suppose the validation test score is higher than the best test accuracy score found so far, and the validation model is reliable. In that case, the best parameters found so far are updated with the ones found in the validation phase. A model is reliable if the training accuracy score exceeds the test accuracy score, which means the model is not underfitting. Moreover, the `bestTestScore` procedure computes the model with the test validation score. The procedure returns three values, the model training accuracy score ($train_s_e$), the test accuracy score ($test_s_e$), and the predictions from the test set ($test_pred_e$). If the test score is higher than the best test accuracy score found so far, and the test model is reliable, the best parameters found so far are updated with the ones found in the test phase. Finally, the best training accuracy score, best test accuracy score, and best predictions are returned from the procedure.

A classification report is obtained for the seven models, including precision in training and test sets. Additionally, a confusion matrix of the best result

achieved in each model is presented, aiding in evaluating each model’s performance by indicating the number of correct and incorrect predictions. This systematic approach aims to maximize the robustness and effectiveness of the classifier ensemble within the context of small and unbalanced datasets.

Algorithm 2 Consensus Meta-Models

Require: Predictions of different models y_pred
 $y_predicted \leftarrow sort(y_pred, by\ accuracyScore)$
 $y_consensus_all \leftarrow majorityVote(y_pred)$
 $y_pred_top \leftarrow getFirst(y_pred, 3)$
 $y_consensus_top \leftarrow majorityVote(y_pred_top)$

In Algorithm 2, the “Consensus” meta-model is defined to obtain the best result from the models’ predictions. For this purpose, the predictions of the seven models are combined with y_pool , and the Consensus Models are applied. The Consensus All Model considers all predictions and selects the most frequent category among the seven models to provide an overall view. On the other hand, the Consensus Top Model relies solely on the three best models, selecting the most frequent category among their predictions and highlighting the influence of the most robust classifiers. The model’s output displays the predictions and the obtained consensus, offering a comprehensive perspective and highlighting the contribution of the most robust models.

4 Results

As discussed in the previous section, a classification report has been obtained for each of the seven models. As discussed in the last section, a classification report has been obtained for each of the seven models. This includes the accuracy of the training set and the test set. In addition, there is also a confusion matrix, which helps to evaluate the performance of the models by indicating the number of correct and wrong predictions [25]. This systematic approach seeks to maximize the robustness and efficiency of the classifier ensemble in the context of small and unbalanced data [26]. Table 1, presents the results of the Accuracy Test, which indicates the proportion of correct predictions made by the model on the test dataset, and Recall 0, which means the proportion of class 0 instances correctly identified by the model relative to all class 0 instances present in the test data, for each of the models used in the “Polisabio Advanced Classifier Optimizer”, as well as the results of the consensus models.

The results reveal that the best overall accuracy and recall of class 0 are achieved with the consensus models, underscoring the effectiveness of this algorithm in obtaining the most fitting and robust model. This is the distinctive feature of the “Polisabio Advanced Classifier Optimizer” through consensus

Model	Accuracy Test	Recall 0
LogReg	0.67	0.80
Decision Trees	0.87	0.60
SVM	0.60	0.60
Naive Bayes	0.67	0.60
Random Forest	0.87	0.80
XG-Boost	0.80	0.80
y_pool	0.87	1.00
Consensus All	0.87	1.00
Consensus Top	0.93	0.80

Table 1. Results of predictive models

meta-models. Figure 3 shows that the output file adds the seven proposed predictions of the models and, in addition, calculates two consensus predictions in Figure 4.

	y_test_svm	y_test_logreg	y_test_naiveB	y_test_xgb	y_test_pool	y_test_trees	y_test_rforest
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
2	0	0	1	0	0	1	1
3	0	0	0	0	0	1	1
4	1	1	1	1	1	1	1
5	1	0	0	1	1	1	1
6	1	0	1	0	0	1	0
7	1	1	0	1	1	1	1
8	1	0	1	1	1	1	1
9	0	1	1	1	1	1	1

Fig. 3. Output: output of predictions of each model.

On the one hand, it calculates “Consensus All”, which considers the outcome predicted by each of the seven models and selects the most frequent category among all predictions, providing a comprehensive view based on the contribution of all models. The “Consensus Top” is based only on the results of the top three models and selects the most frequent category among the predictions of these top three models, in this case, random forest, decision tree, and the best model found in the pool (XGB). This approach considers only the top models’ contributions, highlighting the influence of the most robust classifiers.

y_consensus_all	y_consensus_top	y_test
0	0	0
0	0	0
0	1	1
0	1	1
1	1	1
1	1	1
0	0	0
1	1	1
1	1	1
1	1	1

Fig. 4. Output: output of predictions of each consensus model.

Its comprehensiveness characterizes the developed classification model and is a robust tool, even in a relatively small dataset with only 96 observations [27]. This model has been strategically enhanced to improve performance and enable the Machine Learning algorithms to provide highly accurate results.

In the same way that a detailed evaluation was performed on each classification model, one was generated for each consensus model. This evaluation comprises the accuracy, the confusion matrix, and the classification report. Figure 5 shows the report for the “Consensus All” and “Consensus TOP” model.

The confusion matrix in the Consensus All Model shows that, of the 15 test cases, the classifier correctly predicted 5 cases as negative (true negatives) and eight as positive (true positives). There were two cases where the model incorrectly predicted the negative label (false negatives), resulting in a perfect accuracy for the positive class (1.00) and an accuracy of 0.71 for the negative class.

The recall, also known as sensitivity, for the negative class was perfect (1.00), indicating that all true negative cases were correctly identified. However, the recall for the positive class was 0.80.

The model’s overall accuracy was 0.87, meaning that 87% of the total predictions were correct. The average metrics, macro and weighted, for accuracy, recall, and F1 score were consistently high, with values between 0.86 and 0.90. These results indicate a balanced classifier performance in both classes.

On the other hand, the confusion matrix on the right-hand side of Figure 5, illustrates the confusion matrix for the “Consensus Top” classifier, which achieved a higher test accuracy of 0.93.

The Consensus Top Model accurately predicted 4 cases as class 0 and 10 cases as class 1, with only 1 false positive and no false negatives. In terms of

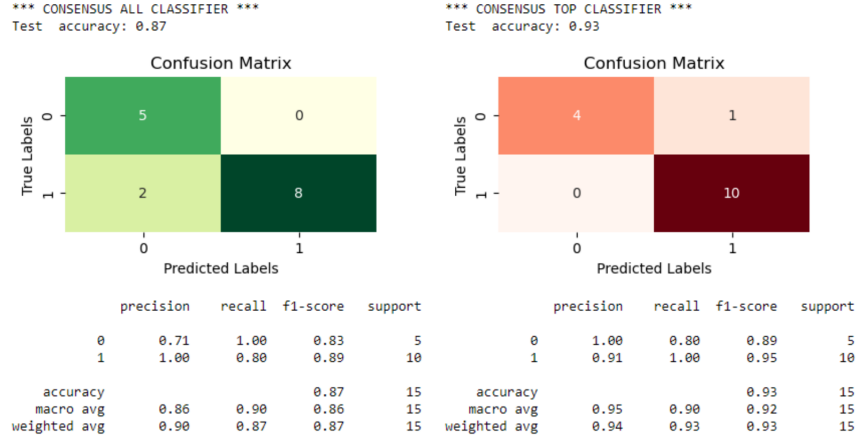


Fig. 5. Consensus All and TOP exit report.

metrics (both individual per class and average metrics), the results are slightly higher than in Consensus All.

The comparison between the “Consensus All” and “Consensus Top” classifiers underlines the relevance of careful model selection in the development of classification systems [28]. Although both methods proved effective, the “Consensus Top” stood out for its superior accuracy, recall, and F1-score performance. This suggests that focusing on the predictions of the most robust models can improve classification system performance.

5 Conclusions

The development of a classification model to predict treatment success in pediatric patients with obesity using a standard treatment represents a significant advancement in personalized healthcare. This approach, grounded in various variables including sociodemographic, familial, habits, and clinical data, demonstrates the potential of artificial intelligence in informed clinical decision-making. Using historical data to train the model and subsequently apply it to new patients establishes a predictive framework that guides treatment selection and optimizes medical resources, considering that a standard treatment does not consume high resources, while a more rigorous and exhaustive one does.

Introducing the “Polisabio Advanced Classifier Optimizer” effectively addresses challenges associated with small and unbalanced datasets. Through techniques such as data oversampling and balancing, this model enhances predictive capability and improves prediction accuracy. Including multiple machine learning algorithms and selecting the best models based on specific criteria further strengthen the system’s robustness.

Consensus models, “Consensus All” and “Consensus Top,” represent a solid approach to integrating predictions from multiple models and improving the reliability of estimates. A comprehensive and accurate view of treatment outcomes is obtained by considering all models’ contributions and focusing on the most robust models.

The results of this study underscore the promise of artificial intelligence in medicine, especially in areas where precision and personalization are crucial, such as pediatric obesity interventions. The high accuracy and effectiveness demonstrated by the model suggest its potential to optimize healthcare resources.

This approach lays the groundwork for future research and clinical applications that leverage the power of artificial intelligence to benefit patient health. Possible research directions could include analyzing data from patients undergoing specific treatments. Thus, we would not only obtain information on patient evolution but could also define which variables contribute most to each treatment and which patient profile is most suitable for each treatment. Furthermore, conducting long-term studies with treatment duration between one or two years would also provide valuable information on the evolution of treatment success.

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References

1. Diana M. Bravo-Saücicela. Has the prevalence of childhood obesity in Spain plateaued? a systematic review and meta-analysis. *International Journal of Environmental Research and Public Health*, 9(5240), 2022.
2. Government of Spain. <https://www.comisionadopobrezainfantil.gob.es/es/plan-estrategico-nacional-para-la-reduccion-de-la-obesidad-infantil>. Last accessed 6 Mar 2024.
3. Ferruccio Pelone, Maria Lucia Specchia, Maria Assunta Veneziano, Silvio Capizzi, Sabina Bucci, Agostino Mancuso, Walter Ricciardi, and AG De Belvis. Economic impact of childhood obesity on health systems: a systematic review. *obesity reviews*, 13(5):431–440, 2012.
4. Mehak Gupta, Thao-Ly T Phan, H Timothy Bunnell, and Rahmatollah Beheshti. Obesity prediction with ehr data: A deep learning approach with interpretable elements. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(3):1–19, 2022.
5. Hao Gou, Huiling Song, Zhiqing Tian, and Yan Liu. Prediction models for children/adolescents with obesity/overweight: A systematic review and meta-analysis. *Preventive Medicine*, page 107823, 2023.
6. Éadaoin M Butler, Avinesh Pillai, Susan MB Morton, Blake M Seers, Caroline G Walker, Kien Ly, El-Shadan Tautolo, Marewa Glover, Rachael W Taylor, Wayne S

- Cutfield, et al. A prediction model for childhood obesity in new zealand. *Scientific reports*, 11(1):6380, 2021.
7. Yuanqing Fu, Wanglong Gou, Wensheng Hu, Yingying Mao, Yunyi Tian, Xinxiu Liang, Yuhong Guan, Tao Huang, Kelei Li, Xiaofei Guo, et al. Integration of an interpretable machine learning algorithm to identify early life risk factors of childhood obesity among preterm infants: a prospective birth cohort. *BMC medicine*, 18(1):1–10, 2020.
 8. Janice L Kwan, Lisha Lo, Jacob Ferguson, Hanna Goldberg, Juan Pablo Diaz-Martinez, George Tomlinson, Jeremy M Grimshaw, and Kaveh G Shojania. Computerised clinical decision support systems and absolute improvements in care: meta-analysis of controlled clinical trials. *Bmj*, 370, 2020.
 9. Benjamin Paassen, Jessica McBroom, Bryn Jeffries, Irena Koprinska, Kalina Yacef, et al. Mapping python programs to vectors using recursive neural encodings. *Journal of Educational Data Mining*, 13(3):1–35, 2021.
 10. Guidelines for the prevention of overweight and obesity in childhood and adolescence. <https://www.unicef.org/media/96096/file/Overweight-Guidance-2020-ES.pdf>. Last accessed 6 Mar 2024.
 11. Spanish Society of Pediatric Gastroenterology, Hepatology and Nutrition (2020). <https://www.seghnp.org/>. Last accessed 6 Mar 2024.
 12. Thomas Reinehr, Nina Lass, Christina Toschke, Juliane Rothermel, Stefanie Lanzinger, and Reinhard W Holl. Which amount of bmi-sds reduction is necessary to improve cardiovascular risk factors in overweight children? *The Journal of Clinical Endocrinology & Metabolism*, 101(8):3171–3179, 2016.
 13. Sebastian Raschka, Joshua Patterson, and Corey Nolet. Machine learning in python: Main developments and technology trends in data science, machine learning, and artificial intelligence. *Information*, 11(4):193, 2020.
 14. Cedric Seger. An investigation of categorical variable encoding techniques in machine learning: binary versus one-hot and feature hashing. (2018:596):34, 2018.
 15. J. Sadaiyandi, P. Arumugam, A.K. Sangaiah, and C. Zhang. Stratified sampling-based deep learning approach to increase prediction accuracy of unbalanced dataset. *Electronics*, 12:4423, 2023.
 16. Mardhiya Hayaty, Siti Muthmainah, and Syed Muhammad Ghufuran. Random and synthetic over-sampling approach to resolve data imbalance in classification. *International Journal of Artificial Intelligence Research*, 4(2):86–94, 2020.
 17. Georgios Douzas, Fernando Bacao, and Felix Last. Improving imbalanced learning through a heuristic oversampling method based on k-means and smote. *Information Sciences*, 465:1–20, 2018.
 18. Michael E Miller, Siu L Hui, and William M Tierney. Validation techniques for logistic regression models. *Statistics in medicine*, 10(8):1213–1226, 1991.
 19. Yan-Yan Song and LU Ying. Decision tree methods: applications for classification and prediction. *Shanghai archives of psychiatry*, 27(2):130, 2015.
 20. Wenjian Wang, Changqian Men, and Weizhen Lu. Online prediction model based on support vector machine. *Neurocomputing*, 71(4-6):550–558, 2008.
 21. Shweta Kharya and Sunita Soni. Weighted naive bayes classifier: a predictive model for breast cancer detection. *International Journal of Computer Applications*, 133(9):32–37, 2016.
 22. Jaime Lynn Speiser, Michael E Miller, Janet Tooze, and Edward Ip. A comparison of random forest variable selection methods for classification prediction modeling. *Expert systems with applications*, 134:93–101, 2019.

23. Pritika Bahad and Preeti Saxena. Study of adaboost and gradient boosting algorithms for predictive analytics. In *International Conference on Intelligent Computing and Smart Communication 2019: Proceedings of ICSC 2019*, pages 235–244. Springer, 2020.
24. Shenglong Li and Xiaojing Zhang. Research on orthopedic auxiliary classification and prediction model based on xgboost algorithm. *Neural Computing and Applications*, 32:1971–1979, 2020.
25. Sankha Subhra Mullick, Shounak Datta, Sourish Gunesh Dhekane, and Swagatam Das. Appropriateness of performance indices for imbalanced data classification: An analysis. *Pattern Recognition*, 102:107197, 2020.
26. Amalia Luque, Alejandro Carrasco, Alejandro Martín, and Ana de Las Heras. The impact of class imbalance in classification performance metrics based on the binary confusion matrix. *Pattern Recognition*, 91:216–231, 2019.
27. P. Xu, X. Ji, M. Li, et al. Small data machine learning in materials science. *npj Computational Materials*, 9:42, 2023.
28. Divya Jain and Vijendra Singh. Feature selection and classification systems for chronic disease prediction: A review. *Egyptian Informatics Journal*, 19(3):179–189, 2018.